

Set-shifting in healthy children and in children with idiopathic or cryptogenic epilepsy

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Set-shifting is an important aspect of attention and regulation of behaviour. The ability of young children to shift set is debated. Set-shifting is usually examined using sorting tasks that are designed for adults and hence are inadequate for children. In this study, an experimental Sorting Task for Children (STC) is administered to 54 healthy children (aged 4 to 10 years) and compared with a widely used adult sorting task, the Modified Card Sorting Test. This experiment shows that preschool children are capable of set-shifting. An extensive error analysis of the STC reveals that perseverations are not predominant. Hence, set-shifting in young children may be characterised by 'trial and error' rather than by being stuck in a mental set, which is suggested to be typical of adults with frontal lobe damage. A pilot study of 10 children with idiopathic or cryptogenic epilepsy shows unstable set-shifting rather than inability to shift set, which is concordant with known problems in regulation of behaviour.

'Mental set' is a classic concept in neuropsychology. It refers to preparatory adjustment for performing a task (Woodworth and Schlossberg 1966). A mental set facilitates responses for which the subject is prepared and inhibits competing responses. Changing from one set to another, called set-shifting, is necessary to allow the subject to adapt to change as the situation requires. For example, if a child has to subtract after adding, s/he has to set-shift. Continuing to add when subtraction is required can be seen as a difficulty in set-shifting and may be called 'perseveration'. The better the set-shifting aptitude, the more flexible the person is at adapting to change.

To detect disturbances in set-shifting, sorting tasks are widely used, such as the Wisconsin Card Sorting Test (WCST) devised by Grant and Berg (1948), introduced to neuropsychology by Milner (1963) and later modified. Currently, the Modified Card Sorting Test (MCST) (Nelson 1976) is preferred, which is used in the present study. The standard stimuli are four different shapes which differ in colour and number. The response cards have one of the three categories – shape, colour, or number – in common with the stimuli, but none is identical to the stimuli. The participant is told to sort the cards according to a certain rule, which s/he must discover using feedback from the examiner. After six consecutive correct responses the participant is asked to find a different rule. Traditionally, set-shifting ability is measured by the number of achieved sorting categories, i.e. shape, colour, and number, and by the proportion of perseverations, defined as the continuation of a previously current but no longer appropriate response (Sandson and Albert 1984).

The findings of several investigations converge at the age of 10 to 12 years, which is the age at which adult performance is achieved (Chelune and Baer 1986). In addition, the errors observed in the performance of younger children resemble those of adults with frontal lobe damage (Chelune and Baer 1986, Kirk and Kelly 1986). These findings have been interpreted as confirming the hypothesis that frontal areas are essentially non-functional until the early teenage years.

In clinical populations of children (Weyandt and Willis 1994) and recently in developmental psychopathology research (for an extensive review see Pennington and Ozonoff 1996), MCST-type sorting tasks have been frequently used. Impaired performance is interpreted as a sign of frontal lobe dysfunction. Before stating that late functioning of the frontal cortex is responsible for deficient set-shifting, the adequacy of MCST-type sorting tasks in the assessment of children must be queried. Attempts to operationalise flexible functioning in children are mainly derived from adult neuropsychological measures with little or no adaptation to take into account the child's developmental level (Welsh et al. 1991).

For three reasons, the MCST is not suitable for assessing set-shifting in young children. Firstly, there is no certainty that children recognise the sorting categories of the MCST. Siegel (1971, 1974) showed that preschool children have difficulty in recognising 'number' as a separate dimension. Because 'number' is one of the sorting categories of the MCST, it is conceivable that the child's errors when performing this task reflect this difficulty rather than lack of ability to shift set. Secondly, developmental psychologists state that an important difference between adults and children lies in the amount of information they can handle (Torgesen 1994). A child's attention span is short (Luciana and Nelson

1998). Some behaviours that are considered to be typical of adult functioning can be revealed in children if this developmental difference is taken into account. Thirdly, in school-age populations, the MCST-type sorting tasks have poor reliability (Pennington et al. 1996). The validity of the measure is questionable because the performance of children with known problems in regulation of behaviour is related more to general cognitive ability than to behaviour problems (Riccio et al. 1994).

In this experiment we tested the hypothesis that children from 4 years of age onwards are capable of set-shifting. The categories to be sorted and the amount of perceptual information and feedback were adapted for children. The sorting categories 'shape' and 'number' were substituted by 'type of animal' and 'facial expression'. In this way, all sorting categories belonged to the conceptual repertory of young children (Church 1961, Gross and Ballif 1991). The amount of perceptual information was reduced by removing the response cards of the previous set and by removing false responses from the response panel. Feedback was adapted by asking the child to name the category after his or her first correct sorting within a category. If a child continued to sort according to a previous rule, the examiner indicated this and asked the child to look for another category.

This paper compared two tasks, the MCST and the sorting task for children (STC), initially in 28 healthy children. However, because the children made errors other than perseverations, 26 further children (total sample size = 54) were studied to elaborate the error analysis across different age groups. Chelune and Baer (1986) found improvement in performance from age 6 to 7 years and from age 8 to 9 years. Therefore, the following age groups were chosen: 4 to 6, 7 to 8, and 9 to 10 years. In addition, transitions in ability to categorise appear to occur between these age boundaries (Mak and Vera 1999). Set-shifting was also examined in a group of 10 children with idiopathic or cryptogenic epilepsy – idiopathic epilepsy is presumed to have a genetic origin; cryptogenic epilepsy occurs in otherwise healthy persons with no clear cause (International League Against Epilepsy 1993). In children with epilepsy, problems in attention or in regulation of behaviour have been described in the presence of normal intelligence and a favourable prognosis of their epilepsies (Jennekens-Schinkel et al. 1987, D'Allessandro et al. 1990, Stores et al. 1992).

Method

PARTICIPANTS

Fifty-four children in age groups 4 to 6 years ($n=19$), 7 to 8 years ($n=18$), and 9 to 10 years ($n=17$) were recruited from two mainstream elementary schools in two medium-sized towns (approximately 60 000 inhabitants) in The Netherlands. Their mean age was 7.2 years and their mean IQ, measured by the Coloured Progressive Matrices (CPM; Raven et al. 1990), was 123 (SD 17.53).

Ten children with idiopathic or cryptogenic epilepsy who were attending mainstream schools, with normal IQs, and an age (mean 7.2 years) in accordance with that of the healthy children were selected (Table I). All children were outpatients of a neurologist from a university or general hospital. IQ scores were determined using the Dutch version of the Wechsler Intelligence Scale for Children – Revised (WISC-R; De Bruyn et al. 1986) or the CPM (Raven et al. 1990).

INSTRUMENTS

The MCST and an experimental sorting task (STC) were used for this study. These tasks are described in the Appendix.

DEPENDENT VARIABLES WHEN COMPARING MCST AND STC

Performances were evaluated by the number of categories achieved and by the number of perseverations and other errors. A category was considered to be achieved when the child performed six correct sortings consecutively. An error was scored as a perseveration if the child persisted with a sorting category despite feedback that the previous sorting was incorrect. Other errors were defined as non-perseverations. For example, if a child had sorted according to 'type of animal' and 'colour' and was then asked to find a different rule, and his/her first response after this instruction was to sort according to 'type of animal', this error was scored as a non-perseveration. After receiving feedback that this response was incorrect, if the child again sorted according to 'type of animal', this was scored as a perseveration.

ELABORATE ERROR ANALYSIS ON STC

The achieved categories were scored as described above. Errors were classified in three overlapping ways: (1) Errors in searching for a category versus errors in maintaining the category. A searching error was any which occurred after the instruction to identify a (new) category and before the first correct response relating to a new category. A maintaining error was any error after the first correct response within a category and before the 6th consecutive correct response. (2) Perseverations, non-perseverations, and uncommon errors. Perseveration was defined as any response relating to a previous category despite feedback that this was incorrect. Uncommon errors were responses which had no connection with any of the categories, for example, when a child sorted a happy yellow rabbit with a sad blue horse. Non-perseverations were all other errors. (3) Not profiting from feedback

Table I: Patient characteristics of children with idiopathic or cryptogenic epilepsy ($n=10$)

Age (y:mo)	Sex	IQ	Epilepsy type ^a	Seizure type ^b	Time between last seizure and testing
5:2	M	104 ^c	LI	SPS + PS-SG	4 mo
6:0	F	130 ^c	LC	CPS + PS-SG	>1 y
6:8	M	97 ^c	LC	CPS	3 mo
6:8	M	100	LC	CPS	>1 y
7:1	M	118	LC	CPS	5 d
7:4	F	106	LI	CPS	4 mo
7:7	M	108	LI	Not classifiable	2½ mo
7:10	M	127	LC	CPS	2 d
8:1	M	108	LC	SPS + PS-SG	5 wk
8:8	M	106	GI	Absences	2 mo

^a According to International League against Epilepsy 1989.

^b According to ILAE 1981.

^c WISC-R, other IQs on Coloured Progressive Matrices; LI, localisation-related idiopathic; LC, localisation-related cryptogenic; GI, generalised idiopathic; SPS, simple partial seizures; PS-SG, partial seizures secondary generalised; CPS, complex partial seizures.

was reflected in 'stay errors' versus 'shift errors'; in other words, 'not staying with a correct category despite positive feedback' versus 'not shifting to the correct category despite negative feedback'.

PROCEDURE

The healthy children were examined at school, individually in a quiet room. The STC was administered first, followed by the MCST. The STC was administered twice. Immediately after completion of trial 1 (either after the 24th response or after six consecutive correct responses in the three categories) the child was given the second pack of shuffled cards and the procedure was repeated (trial 2). To avoid a loss of motivation, the MCST was administered once. The CPM was then administered. For the children with epilepsy, the administration of the STC was part of a comprehensive neuropsychological assessment.

STATISTICAL ANALYSES

Data were analysed using the Statistical Package for Social Sciences, version 9.0. Because of the skewness of the distrib-

ution of the results, the small groups, and the limited range of scores for the variable 'categories', non-parametric measures were used.

Performance differences across the three age groups were studied with a Kruskal–Wallis one-way analysis of variance (ANOVA). For comparing performances between two age groups, a Mann–Whitney *U* test was used. Within-age group variables were analysed using a Wilcoxon test for paired associates. For all analyses the level of statistical significance was set at 0.05 two-sided. If a *p* value was between 0.05 and 0.10, a tendency to significance was presumed and will be reported.

Results

PERFORMANCE ON MCST AND STC IN 28 HEALTHY CHILDREN FOR EACH AGE GROUP

Because the MCST was administered once, the results were compared with the first trial on the STC.

Number of achieved categories (Table II)

Most of the youngest children (aged 4 to 6 years) achieved

Table II: Comparison of Modified Card Sorting Test (MCST) and Sorting Task for Children (STC) in 28 healthy children: means (SDs) of achieved categories, perseverations, and non-perseverations in three age groups and total group

	Age 4–6 y (n=10)			Age 7–8 y (n=9)			Age 9–10 y (n=9)			All (n=28)			Age effects ^a	
	MCST	STC	<i>p</i> ^b	MCST	STC	<i>p</i> ^b	MCST	STC	<i>p</i> ^b	MCST	STC	<i>p</i> ^b	MCST	STC
Categories (max 3)	1.40 (0.70)	2.90 (0.32)	0.00	2.33 (0.87)	3.00 (0)	0.06	2.89 (0.33)	3.00 (0)	ns	2.18 (0.90)	2.96 (0.19)	0.00	0.00	ns
Perseverations	8.30 (6.41)	0.80 (2.53)	0.01	5.10 (6.47) ^c	0 (0)	0.07	0.11 (0.33) ^d	0 (0)	ns	4.64 (6.17)	0.29 (1.51)	0.00	0.01	ns
Non-perseverations	4.30 (4.03)	1.40 (0.70)	0.09	1.20 (1.48) ^c	0.56 (0.53)	ns	1.89 (1.17) ^d	0.11 (0.33)	0.01	2.54 (2.89)	0.75 (1.04)	0.00	0.08	0.01

^a Kruskal–Wallis ANOVA.

^b Wilcoxon signed rank test.

^c Trend for perseverations > non-perseverations on MCST (*p*=0.08) (Wilcoxon).

^d Non-perseverations > perseverations (*p*=0.01) (Wilcoxon).

Table III: Performance on two trials of the Sorting Task for Children (STC) by 54 healthy children: means (SDs) of achieved categories and errors in the age groups and in the total group

	Age 4–6 y (n=19)	Age 7–8 y (n=18)	Age 9–10 y (n=17)	All (n=54)	Age effects <i>p</i>
	Mean (SD)	Mean (SD)	Mean (SD)	Mean (SD)	
Achieved categories (max 6)	5.74 (0.56)	6.00 (0)	6.00 (0)	5.91 (0.35)	0.02 ^a
Searching errors	3.16 (4.49) ^b	0.33 (0.69)	0.47 (1.46)	0.47 (1.46) ^g	0.00 ^a
Maintaining errors	0.84 (1.17)	0.72 (0.83)	0.29 (0.59)	0.29 (0.59)	ns
Perseverations	0.95 (3.89)	0 (0)	0 (0)	0.33 (2.31)	ns
Non-perseverations	2.68 (2.14) ^c	0.89 (0.96) ^c	0.47 (1.07) ^c	1.39 (1.77) ^c	0.00 ^a
Uncommon errors	0.42 (0.69)	0.11 (0.47)	0.29 (0.85)	0.28 (0.68)	ns
Shift errors	1.58 (3.85)	0.06 (0.24)	0 (0)	0.57 (2.37)	0.00 ^a
Stay errors	0.84 (1.17)	0.72 (0.83) ^d	0.53 (1.28) ^f	0.70 (1.09) ^f	ns
Total error	4.00 (4.94)	1.06 (1.26)	0.76 (1.75)	2.00 (3.46)	0.00 ^a

^a Significant for Kruskal–Wallis one-way ANOVA.

For Wilcoxon matched pairs: ^b Searching errors > maintaining errors (*p*<0.01); ^c Non-perseverations > perseverations and larger than uncommon errors (*p*<0.01); ^d Stay errors > shift errors (*p*<0.01); ^e Non-perseverations > perseverations and larger than uncommon errors (trend, *p*=0.06); ^f Stay errors > shift errors (trend, *p*=0.06);

^g Searching errors > maintaining errors (trend, *p*=0.09).

only one sorting, but could not set-shift on the MCST. On the STC, however, nearly all preschool children achieved three categories. The 7- to 8-year-old children performed better than the youngest children, but still had difficulties in set-shifting on the MCST. Only five out of nine children in this age group achieved all categories on the MCST, compared with all children on the STC, a difference tending to statistical significance ($p=0.06$). Most of the oldest children (aged 9 to 10 years) were able to shift set even on the MCST.

Errors (Table II)

The youngest children made a lot of perseverations and other errors (non-perseverations) on the MCST. On the STC, however, only one child of this age group made perseverations and the number of non-perseverations also tended to be lower. The 7- to 8-year-old children made up to 17 errors of perseveration on the MCST, compared with none on the STC. In this age group, the non-perseveration errors made on the MCST dropped to four, not significantly different from those on the STC. In the 9- to 10-year-old children, the errors of perseveration were negligible in both sorting tasks, but these children made statistically significantly more non-perseveration errors on the MCST than on the STC.

As far as the proportion of error types is concerned in the STC, the group as a whole made more non-perseverations than perseverations ($p=0.01$). In the MCST this difference did not exist. The two youngest age groups made more perseverations than non-perseverations, a difference that is, however, not significant. The oldest age group (aged 9 to 10 years) made significantly more non-perseverations than perseverations.

ANALYSIS OF PERFORMANCE OF 54 HEALTHY CHILDREN ON STC (TABLE III)

All but one child achieved three categories within 24 responses in trial 1 as well as in trial 2 of the STC. Differences between trial 1 and 2 were a lower total error score and a lower number of non-perseverations, uncommon errors, and searching errors in trial 2 than in trial 1. The youngest children (aged 4 to 6 years) benefited most from repeating the procedure. As in the group as a whole, the pattern of errors in trial 1 and 2 was the same. The scores are summed over the two trials.

Error types within the three error classifications were also compared. Overall, the children tended to make slightly more searching errors than maintaining errors, and slightly more stay than shift errors. However, perseverations, non-perseverations, and uncommon errors revealed statistically significant differences. The children made mainly non-perseverations, fewer uncommon errors, and no perseverations.

Age-related differences in error types (Table III)

Age effects were present in achieved categories, searching errors, non-perseverations, shift errors, and the total error score, i.e. younger children (aged 4 to 6 years) had more difficulties in searching for a category and gained less from feedback. Perseverations were absent and nearly all children managed to achieve the three categories within 24 responses.

PERFORMANCE OF CHILDREN WITH EPILEPSY COMPARED WITH HEALTHY CHILDREN ON STC (TABLE IV)

Within the group of children with epilepsy, non-persevera-

tions outnumbered perseverations and uncommon errors. Uncommon errors tended to outnumber perseverations. There were no other differences between error types; there was no relation with epilepsy type, seizure type, or seizure-free interval.

Compared with the group of 54 healthy children, children with epilepsy achieved fewer categories and made more shift errors. They tended to make more searching errors, non-perseverations, and uncommon errors, and to have a higher total error score than healthy children. These differences appeared mainly in the second trial. In trial 1, children with epilepsy and healthy peers could not be distinguished, but the performances of the two groups differed in trial 2. Rather than an amelioration – as found in the healthy children – we found a subtle deterioration in performance in children with epilepsy.

Discussion

The ability of young children to shift set is debated. The results of this study show that healthy children from the age of 4 years are capable of set-shifting, provided tasks are adapted to their developmental level. Espy (1997), in a different context, demonstrated set-shifting skills in children as young as 5 years of age. In addition, as reported in Smith et al. (1992), Diamond and Boyer (1989), using several experimental versions of sorting tasks, demonstrated that preschool children have the cognitive skills necessary for set-shifting. Their experiments have not been published extensively, but their conclusions appear to be similar to ours.

Using the WCST designed for adults, Chelune and Baer (1986) conclude that set-shifting in children is poor. Luciana and Nelson (1998) report on the Cambridge Neuropsychological Test Automated Battery (CANTAB; also designed for adults) in 4- to 8-year-old healthy children; 4-year-old children could shift. However, shifting between two categories – comparable with the WCST – was too difficult, but this improved with age. Eight-year-old children

Table IV: Performance on two trials of the STC of healthy children ($n=54$) and children with epilepsy ($n=10$): means (SDs) of achieved categories and errors

	Healthy children ($n=54$) Mean (SD)	Children with epilepsy ($n=10$) Mean (SD)	Comparison p
Achieved categories (max 6)	5.90 (0.35)	5.50 (0.71)	0.00 ^a
Searching errors	1.37 (3.07)	2.60 (3.31)	0.07
Maintaining errors	0.63 (0.92)	1.70 (2.06)	0.10
Perseverations	0.33 (2.31)	0.10 (0.32)	ns
Non-perseverations	1.39 (1.77)	3.00 (3.37) ^b	0.08
Uncommon errors	0.28 (0.68)	1.20 (1.87) ^c	0.06
Shift errors	0.57 (2.37)	1.10 (1.66)	0.03 ^a
Stay errors	0.70 (1.09)	1.00 (1.94)	ns
Total errors	2.00 (3.46)	4.30 (5.12)	0.08

^a Significant for Mann–Whitney U test.

^b Non-perseverations > perseverations ($p=0.01$), non-perseverations > uncommon errors ($p=0.02$).

^c Trend for uncommon > perseverations ($p=0.07$).

performed similar to adults. When comparing different tasks designed to assess regulation of behaviour in school-children, Weyandt and Willis (1994) found the age at which adult performance is achieved varies across tasks. Their results, like ours, demonstrate the need for developmentally appropriate measures.

When comparing the performances on the STC and the MCST, we found the difference to be largest in the youngest group. The 4 to 6 year olds gained most from the adaptations of the STC. One might object that the adaptations changed the sorting task in such a way as to render set-shifting almost unavoidable. We strongly deny this. Efficient processing of a task is significantly influenced by the child's knowledge of and familiarity with the type of information being processed (Torgesen 1994). We therefore adapted sorting categories. As nearly all children identified all categories, we confirmed that these belonged to the conceptual repertory of the child. Our further adaptations of feedback certainly facilitated responses. It should be stressed that the child discovered the new category, made the shift and named the category after identifying it. The STC did not require mere matching, but remained a task of set-shifting. Our result of early set-shifting skills is in accordance with that of Luciana and Nelson 1998. In this study, as in the STC, the responses relating to the previous category were removed. This condition presumably facilitated the demonstration of set-shifting capabilities at an earlier age.

In adults, perseverations are considered to be the characteristic error type when set-shifting is hampered by brain damage. Usually, dysfunction of frontal lobe areas is considered when shifting behaviour is compromised, and, in children, insufficient or absent shifting of set is explained by immaturity of these frontal structures.

However, perseverations do not appear to be the characteristic feature of children's shifting behaviour. In both the MCST and the STC, the children made other errors, but there was not even a predominance of perseverations. Overall, younger children achieved fewer categories, made more searching errors, shift errors and non-perseverations than older children, but not more perseverations. This might imply that younger children sort by 'trial and error'. Being 'stuck in a mental set', which is thought to be characteristic of adults with frontal dysfunction (Milner 1963, Heaton 1981), is not typical of children.

In addition, in the children with epilepsy, perseverations were not typical. They achieved fewer categories and made more shift errors than healthy children. Perseverations were even fewer than in healthy children although this difference was not significant. As the children with epilepsy could not be distinguished from the healthy children in trial 1 of the STC, set-shifting in itself seemed not to be disturbed. In trial 2, however, their performance decreased whereas in the healthy control children a slight improvement was seen. Although this preliminary result does not allow one to draw conclusions on set-shifting in children with epilepsy, the finding supports the notion of instability in those regulatory or attentional processes of which set-shifting is an example. This is concordant with findings in children with idiopathic or cryptogenic epilepsy, as shown by tests of attention (D'Allessandro et al. 1990, Stores et al. 1992) and writing errors made by children with normal intelligence and mild epilepsy (Jennekens-Schinkel et al. 1987).

In conclusion, children are capable of set-shifting from the age of 4 years. Perseverations are not the predominant error type. The error patterns found in younger children and in children with idiopathic or cryptogenic epilepsy suggest that not 'being stuck in a mental set' but a 'trial-and-error-approach' is at the root of immature and deficient shifting behaviour.

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Appendix

DESCRIPTION OF METHOD

The Modified Card Sorting Test (Nelson 1976)

While holding a pack of cards, the examiner says: 'You will see triangles, stars, crosses, or circles on these cards. Each card has one, two, three or four of them. They might be red, green, blue, or yellow.' While pointing to the standard stimuli, the examiner says: 'Please sort the cards according to these examples. You have to sort them according to a certain rule, but I am not going to tell you what the rule is. You have to find that out for yourself. When you put down a card, I will tell you whether it is right or wrong. If it is wrong, leave the card where it is and try again.' After six consecutive correct responses, the examiner says: 'Now we are going to change the rule; you have to find the new rule by trial and error. Again, I'll tell you whether the card you put down is right or wrong.' After six consecutive correct responses, the examiner repeats the instruction for the third sorting. The task is concluded when three categories have been achieved or when 24 sortings have been made.

The Sorting Task for Children (STC)

The STC is similar to the MCST, with modifications to render the task suitable for children. The material consists of a panel with the four standard stimuli and two identical packs of response tokens, animal shapes cut out of foam rubber. The standard stimuli are four types of animal (horses, whales, squirrels, and rabbits), each presented in one of four colours (yellow, blue, red, or green) and provided with one of four facial expressions (happy, sad, angry, or no expression at all). Each of the response tokens has one feature (type of animal, colour, or facial expression) in common with the standard tokens. The child uses one pack of 24 shuffled response tokens.

While pointing to the standard stimuli, the examiner says: 'Here is a yellow horse which is sad, a blue whale which is happy, a red squirrel which is angry, and a green rabbit, we do not know how it feels. In my hand I have more horses, whales, squirrels, and rabbits. They might be yellow, blue, red, or green. Some are happy, some are sad, some are angry, and in some cases we cannot see how the animal feels.' While pointing to the response panel, the examiner says: 'You can place the animals I give you here.' The examiner gives the first response token to the child and asks: 'Where does this one belong?' The first response that reflects the application of one of the three adequate sorting rules (type of animal, colour, or facial expression) is reinforced by asking: 'These belong together because...?' The child names the category. Incorrect responses are removed and the child is encouraged to try again.

After six correct responses, the examiner removes and shuffles the response tokens, while instructing the child to sort according to

a different rule, by saying: 'Now we are going to do this in another way.' The examiner gives the first token while saying: 'In what other way are they also alike? Where does this one belong?' When the child responds correctly by sorting according to a different rule the examiner reinforces this by saying 'Yes, that's right, these belong together because...?' The child names the category. When the child continues to sort according to the previous rule the examiner removes the incorrect response and says: 'We've already done it this way, try to find another way. How else are they alike?' When a child persists with the previous category the examiner says: 'You have already put (for example) the colours together. In what other way do they fit together?' Incorrect responses are removed.

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